CS-302: Artificial Intelligence

LECTURE NO 4:

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Organization of Lecture

1. AI Technique
2. Parts of AI Technique
3. Uninformed Search Strategies
    1. Breadth First Search (BFS)
    2. Depth First Search (DFS)

1. AI Technique

1) AI Technique: It is a method that exploits knowledge that should be represented in such a way that:

i) knowledge captures generalization
ii) Understandable by people
iii) Can easily be modified to correct errors.
iv) can be used in many situations.
v) can reduce its volume on its own (for example knowledge no more required should be minimized or reduced by AI technique itself).

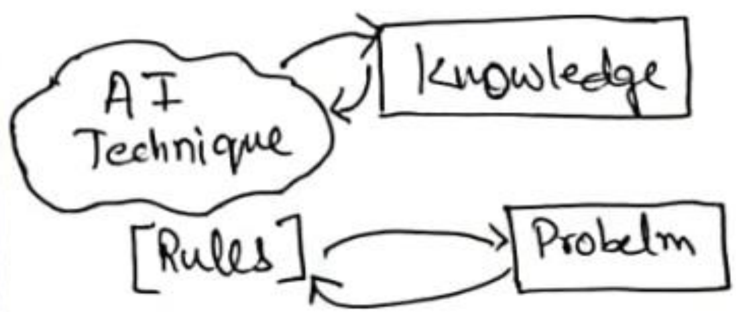
2. Parts of AI Technique

2) Parts of AI Technique:-

① **Knowledge Representation:** It is used to capture the knowledge about the real world. (knowledge gathering from real world).

② **Search Algorithm:** Finding / Searching solution of the probelm.

Knowledge
→ Large in Volume.
→ Not well formatted.
→ Constantly Changing.



3. Uninformed Search Strategies (1/9)

③ Uninformed Search

3.1 BFS: Breadth First Search

↳ Explores all the nodes at given depth before proceeding to the next level.

↳ Uses Queue to implement.

Algorithm:

i) Enter starting nodes on Queue.

ii) If Queue is empty, then return fail & stop.

iii) If first element on Queue is GOAL Node, then return Success & stop.

ELSE

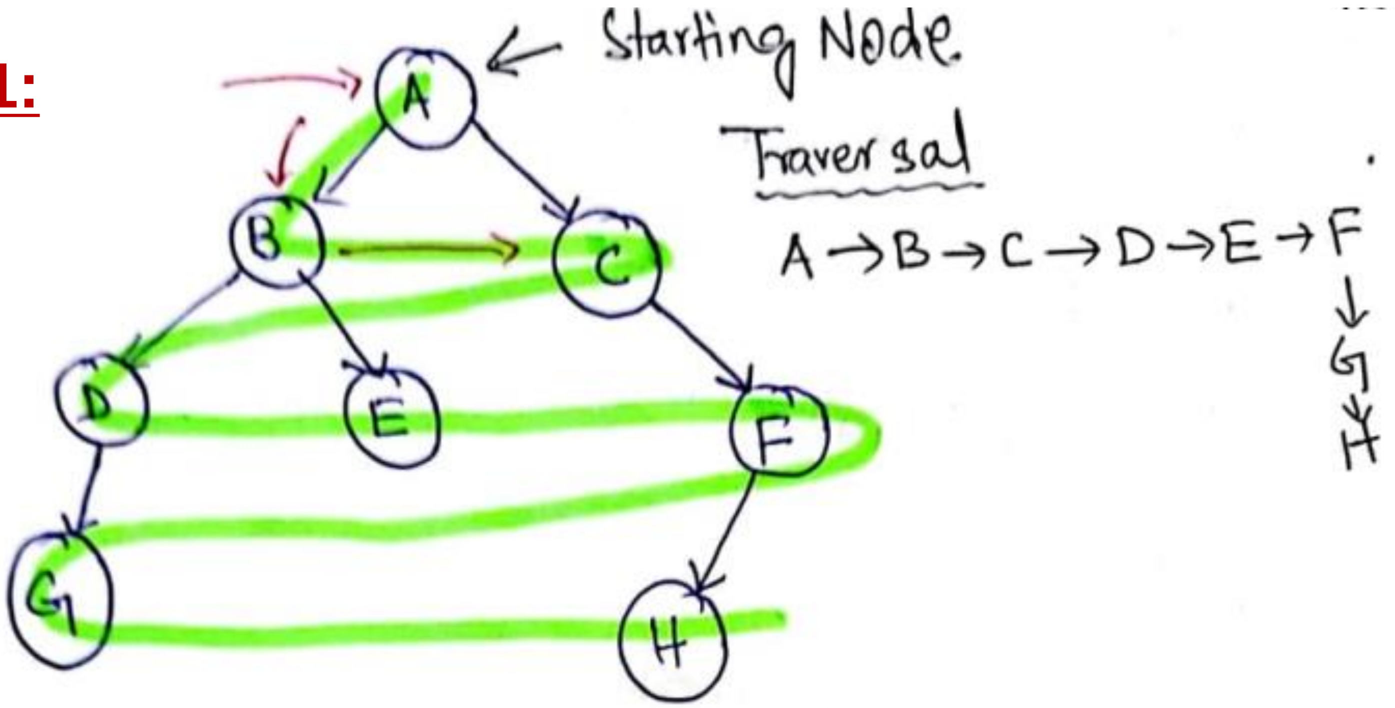
IMP STEP

iv) [Remove and expand first element from Queue and place chidern at the end of Queue.]

v) Goto step (ii)

3. Uninformed Search Strategies (2/9)

Example 1:



Initial Queue {A} = Goal node X.
↳ Remove from Queue and add its successor to Queue.

3. Uninformed Search Strategies (3/9)

→ Successors of A are B & C.

{B, C}

→ Not Goal state/node, Remove Node B & expand it by placing its childernnodes (D&E) at the end of Queue.

{C, D, E}

→ Remove C as it is not the Goalnode & same step as above.

{D, E, F}

→ Removed D & same step as above.

{E, F, G}

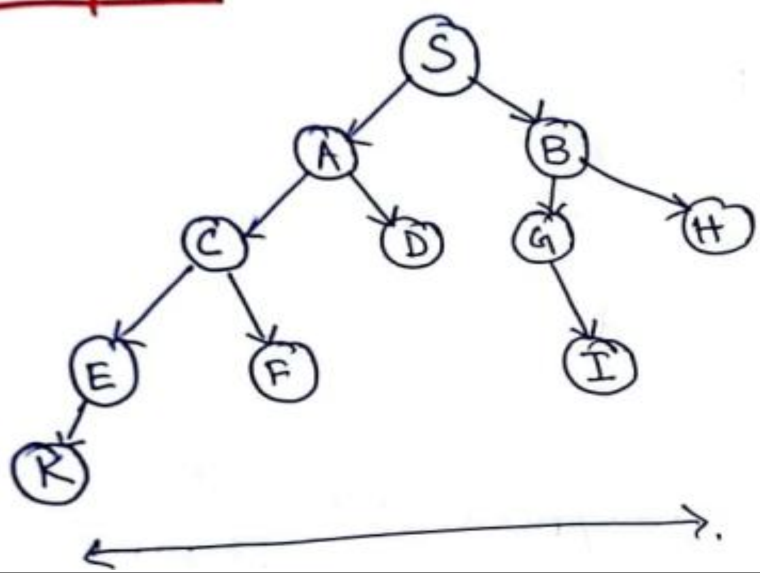
3. Uninformed Search Strategies (4/9)

{E, F, G}
→ Remove E

{F, G}
→ Remove F

{G, H}

Example 2: Home Work



3. Uninformed Search Strategies (5/9)

Advantages: i) BFS will find a solution if it exists.
ii) It will try to find the minimal solution in least no. of steps.

Disadvantages:
i) It requires more memory (b/c tree is expandedat every stage)
ii) Needs lots of time if solution is far from root node.

Time Complexity (T.C) & Space Complexity (S.C)

↳ $O(b^d)$

O → order of

b → branching factor

d → depth

3. Uninformed Search Strategies (6/9)

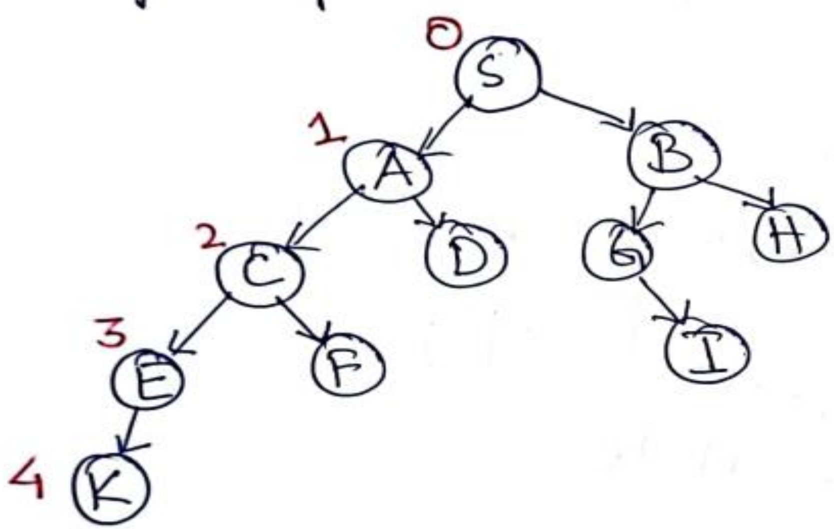
Where b= branching factor (How many times a node is dividing?)

→ for example Node S in example 2 is divided into two nodeA & node B.

therefore b=2

d = depth.

→ for example depth of tree in example 2 is 4 i-e



3. Uninformed Search Strategies (7/9)

3.2 **DFS:** Depth-First Search.

↳ Recursive Algorithm

↳ Starts from root node and follows each path to its greatest depth node before moving to the next path

↳ implemented using STACK. [LIFO]

Algorithm:

i) Enter ROOT (START) node on STACK. (PUSH)

ii) Do until Stack is not empty

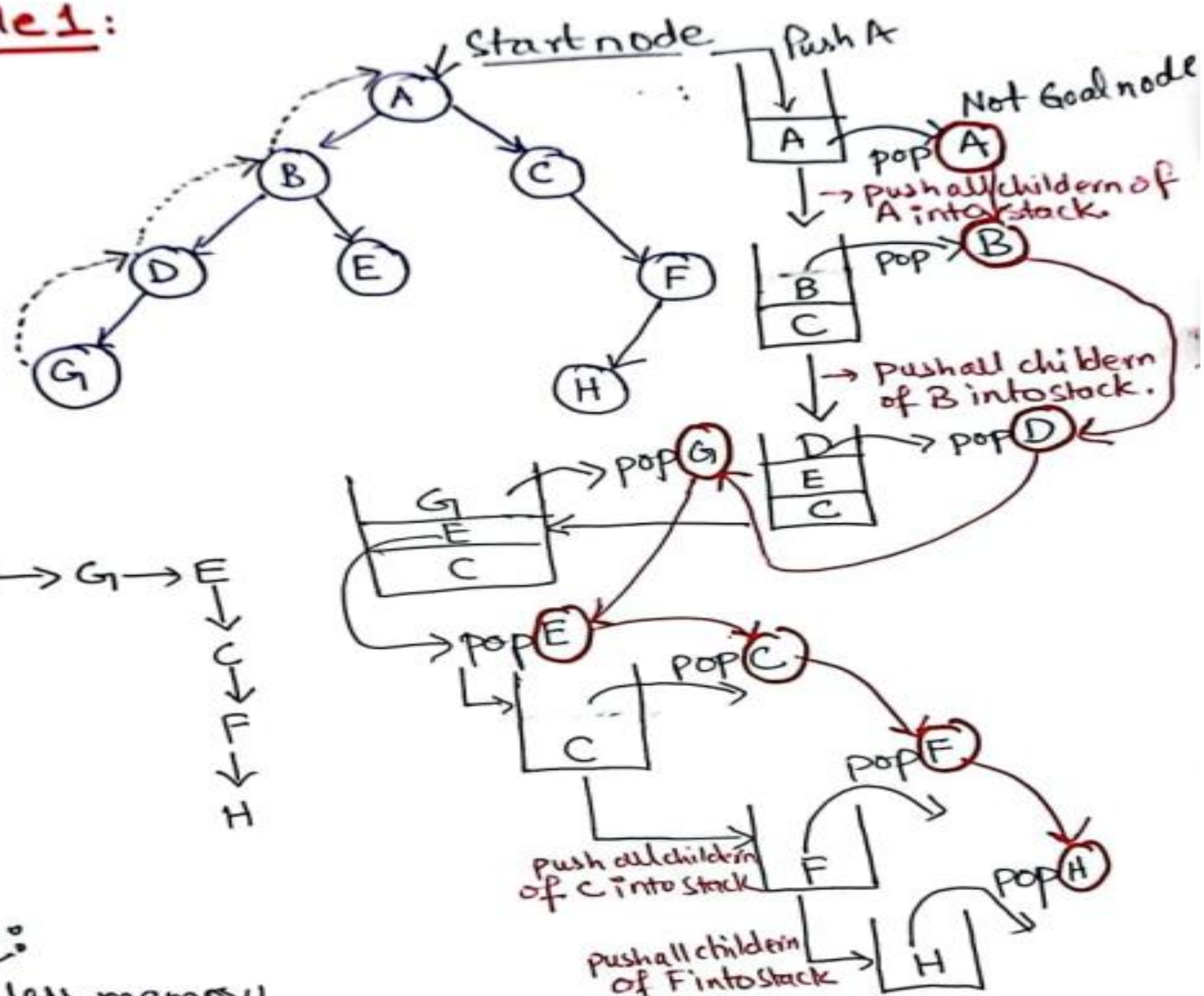
↳ a) Remove Node (POP)

↳ i) If Node = Goal, stop

ii) Push all childern of node in Stack.

3. Uninformed Search Strategies (8/9)

Example 1:



Traversal

A → B → D → G → E → C → F → H

Advantages:

i) Require less memory

ii) Less time to reach goal node if traversal in right path.

3. Uninformed Search Strategies (9/9)

Disadvantages:

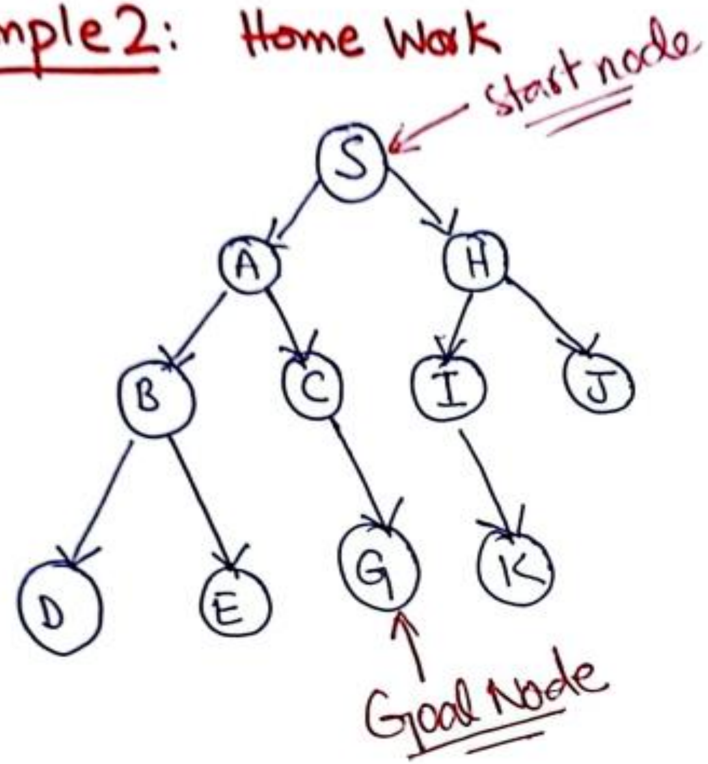
i) No guarantee of finding a solution.

ii) It can go in infinite loop.

Time Complexity = $O(b^d)$ ⇒ same as BFS

Space Complexity = $O(bd)$.

Example 2: Home Work



References: Textbooks

1. *"Artificial Intelligence", by Elaine Rich and Kevin Knight, (2006), McGraw Hill companies Inc., Chapter 1-22 page 1-613.*
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3. *"Computational Intelligence: A logical Approach", by David Poole, Alan Mackworth, and Randy Goebel, (1998), Oxford University Press, Chapter 1-12, page 1-608.*
4. *"Artificial Intelligence: Structures and Strategies for Complex Problem Solving", by George F.Lugar, (2002), Addison-Wesley, Chapter 1-16, page 1-743.*
5. *"AI: A new Synthesis", by Nils J.Nilsson, (1998), Morgan Kaufmann Inc., Chapter 1-25, Page 1-493.*
6. *"Artificial Intelligence: Theory and Practice" by Thomas Dean, (1994), Addison-Wesley, Chapter 1-10, Page 1-650.*
7. *Related documents from open source, mainly internet.*